fish like coelomate forms with chordate characters such us notochord, dorsal tubular nerve cord and pharyngeal gill slits throughout their life. Closed type of circulatory system is seen without heart. Excretion is by **protonephridia**. Sexes are separate, Fertilization is external. Development is indirect and includes a free swimming larva (Figure 2.21).

Example: *Branchiostoma* (Amphioxus or lancelet)

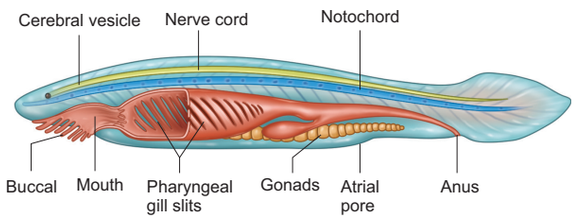


Figure 2.21 Example of Cephalochordata - Amphioxus

2.4.3 Subphylum-Vertebrata

(L. *Vertebrus* –back bone)

Vertebrates are also called higher chordates and they possess notochord during embryonic stage only. The notochord is replaced by a cartilaginous or bony vertebral column in the adult. **Hence all vertebrates are chordates but all chordates are not vertebrates.** Vertebrates possess paired appendages such as fins or limbs. Skin is covered by protective skeleton comprising of scales, feathers, hairs, claws, nails, etc. Respiration is aerobic through gills, skin, buccopharyngeal cavity and lungs. Vertebrates have a ventral muscular heart with two, three or four chambers and kidneys for excretion and osmoregulation.

Subphylum Vertebrata is divided into two divisions, **Agnatha** and **Gnathostomata**. Agnatha includes jawless fish-like aquatic vertebrates without paired appendages. Notochord persists in the adult. Gnathostomata includes jawed vertebrates with paired appendages. Notochord is replaced partly or wholly by the vertebral column. Agnatha includes one important class – Cyclostomata. **Gnathostomata** includes **jawed fishes** (Pisces) and **Tetrapoda** (amphibia, reptilia, aves and mammals). The superclass Pisces includes all fishes which are essentially aquatic forms with paired fins for swimming and gills for respiration. Pisces includes cartilaginous fishes (Chondrichthyes) and bony fishes (Osteichthyes).

2.4.4 Class: Cyclostomata

(G.*cyklos*–circle; *stomata* -mouth)

All members of cyclostomata are primitive, poikilothermic, jawless aquatic vertebrates and are ectoparasites on some fishes. Body is slender and eel-like bearing six to fifteen pair of gill slits for respiration**.** Mouth is circular without jaws and suctorial. Heart is two chambered and circulation is of closed type. No paired appendages. Cranium and vertebral column are cartilaginous. Cyclostomes are marine but migrate to fresh waters for spawning **(anadromous migration)**. After spawning within a few days they die. The larvae **(ammocoete)** after metamorphosis returns to the ocean. Examples: *Petromyzon* (Lamprey) and *Myxine* (Hag fish) (Figure 2.22).



Lamprey



Hag fish

Figure 2.22 Examples of Cyclostomata

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